

What does language grounding actually contribute to cell representations?

A controlled decomposition on held-out donors

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Abstract

Language-grounded single-cell models are hypothesized to produce better, more transferable representations than purely molecular models, but the benefit is rarely decomposed: *which part* of the grounding text carries the transferable signal? We built a controlled benchmark on 85,055 human blood cells (Tabula Sapiens, 9 donors, 16 cell types) that compares a supervised cell-only encoder against CLIP-style contrastive cell-text alignment across four grounding sources — per-cell gene lists, train-derived marker lists, LLM-generated biological prose written from markers alone, and leakage probes — with two frozen text encoders, all evaluated on held-out donors. We find that zero-shot classification from marker-list grounding matches a fully supervised baseline (0.591 vs. 0.603 macro-F1; balanced accuracy 0.635 vs. 0.618) without any label-trained weights, while *cell-internal* label-free sources remain substantially weaker (0.29–0.40). Swapping a general sentence encoder for a biomedical entity-linking encoder changes little; the grounding *source* is the dominant factor. We then test whether *external* knowledge closes the gap: PubMed text retrieved by marker queries over a 30.8M-article corpus, with cell-type names stripped, reaches 0.563 macro-F1 — 95% of the label-derived arm — and a fully label-free pipeline (curated markers → retrieval → LLM synthesis) reaches 0.603, matching supervised with no labels anywhere in the chain. A register control proves decisive: the same captions scored against marker-list class captions collapse to 0.15–0.19 F1, a ~ 0.4 penalty attributable to caption format alone. We conclude that text is a viable carrier of transferable biological structure — including structure that does not derive from the training labels — provided that retrieval quality and caption register are controlled. All code, manifests, prompts, retrieved provenance, and generated text are released openly.

1 Introduction

Single-cell foundation models — scGPT [Cui et al., 2024], Geneformer [Theodoris et al., 2023], scFoundation [Hao et al., 2024] and successors — learn cell representations from expression profiles alone. A parallel line of work argues that grounding those representations in biological language unlocks generalization that molecular data alone cannot provide: CellWhisperer aligns transcriptomes with AI-curated textual annotations [Schaefer et al., 2025], C2S-Scale renders expression profiles as “cell sentences” inside a general-purpose LLM [Rizvi et al., 2025], and ongoing open efforts (e.g., the Allen Institute’s CellOLMo direction on top of OLMo [Groeneveld et al., 2024]) aim to build multimodal models where text is a first-class modality rather than an annotation.

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The motivating question — *does language grounding improve cell representations?* — is harder to answer than it looks, because “grounding” bundles at least three distinct mechanisms: (i) text can carry *label-discriminative structure* (captions derived from cell-type labels); (ii) text can carry *external biological knowledge* (function, lineage, marker semantics); and (iii) text can act as a *representation bottleneck* that smooths per-cell measurement noise. Published systems typically report end-to-end performance and leave the decomposition implicit.

Here we isolate these mechanisms with a deliberately small, controlled experiment. On 85,055 human blood cells from Tabula Sapiens [Tabula Sapiens Consortium et al., 2022] accessed via CZ CELLxGENE Census [CZ CELLxGENE, 2023], we train the same cell encoder either end-to-end with labels (supervised baseline) or by contrastive alignment to four different caption sources, and evaluate all models on *whole held-out donors* with no label access at inference. Our contributions:

1. A reproducible benchmark protocol for language-grounded cell representations with donor-level held-out evaluation and explicit leakage probes.
2. Evidence that zero-shot cell-type assignment via marker-list grounding reaches supervised performance on unseen donors (0.591 vs. 0.603 macro-F1), showing that contrastive text alignment can carry class structure across individuals.
3. A decomposition showing this benefit is carried almost entirely by label-derived discriminative text: label-free grounding sources (per-cell gene lists, LLM-inferred prose) recover only 48–66% of the marker-list result, and a biomedical text encoder offers no advantage on gene-list captions.
4. A cautionary engineering finding: identifier-mapping errors (Ensembl/positional feature IDs in place of gene symbols) silently destroy grounding semantics — a failure mode that produced measurable metric differences and is now pinned by regression tests.

2 Related work

Expression-only single-cell foundation models. scGPT [Cui et al., 2024], Geneformer [Theodoris et al., 2023], scFoundation [Hao et al., 2024], and UCE [Rosen et al., 2023] learn cell representations from expression alone; UCE additionally targets cross-species universality via protein-embedding gene tokens. Critically, independent zero-shot evaluations find these models underperform simple baselines [Kedzierska et al., 2025, Atti and Subramaniam, 2025]: Kedzierska et al. [2025] show Geneformer and scGPT lose to HVG selection plus Harmony/scVI on cell-type clustering, and Atti and Subramaniam [2025] report Seurat v5 beating all three on classification. The burden of proof for any added modality is therefore high — motivating the controlled, leakage-aware protocol used here.

Language-grounded cell models. LangCell [Zhao et al., 2024] established contrastive cell–text pretraining on metadata-derived captions; CellWhisperer [Schaefer et al., 2025] scales CLIP-style alignment to $\sim 1\text{M}$ cells with AI-curated descriptions and adds a chat interface; C2S-Scale [Rizvi et al., 2025] and Cell2Sentence [Levine et al., 2024] instead serialize expression as “cell sentences” inside general LLMs. None of these decompose which part of the text carries the transferable signal — our central question.

Knowledge-grounded approaches. At the gene level, GenePT [Chen and Zou, 2024] and scGenePT [Istrate et al., 2024] show text-derived gene embeddings (NCBI/UniProt/GO) are additive but not sufficient alone — consistent with our caption-level finding. Ontology-structured supervision appears in OnClass [Wang et al., 2021], scCello [Yuan et al., 2024], and KCFM [KCFM Team, 2025]; retrieval-based annotation in scRAG [Yu et al., 2025] and scHilda [scHilda Team, 2025]. These ground in *external* knowledge — the direction our results identify as the open gap — but target annotation rather than representation transfer across held-out individuals.

3 Methods

3.1 Data and evaluation protocol

We queried the CZ CELLxGENE Census (version 2025-11-08) for Tabula Sapiens primary blood data, obtaining 85,055 cells from 9 donors annotated into 16 cell types, restricted to the 2,000 most highly variable genes after standard normalization (library-size normalization to 10^4 counts, $\log_1 p$). Crucially, census feature identifiers were mapped to HGNC gene symbols via `feature_name` before any caption construction — positional or Ensembl identifiers carry no lexical semantics for a text encoder (Section 4.6).

Held-out donors. For each of three seeds we split cells by donor (25% of donors held out, whole donors only). Marker tables are computed on training donors only; test cells never contribute labels.

Models. A shared MLP cell encoder ($2000 \rightarrow 256 \rightarrow 128 \rightarrow 64$) produces a 64-d cell embedding. Text is encoded by a frozen sentence model: either general-purpose `all-MiniLM-L6-v2` (Sentence-BERT [Reimers and Gurevych, 2019]) or biomedical `SapBERT-from-PubMedBERT-fulltext` [Liu et al., 2021], mean-pooled. For the grounded arm, a learned linear projection maps the cell embedding to the text-embedding space and training optimizes a symmetric InfoNCE objective [van den Oord et al., 2018, Radford et al., 2021] (temperature 0.1, AdamW, 40 epochs, batch 512).

Arms. (a) *cell-only supervised*: same encoder plus a linear head, trained end-to-end with cross-entropy on labels. (b) *grounded*: contrastive alignment against per-cell captions, then evaluated two ways — *zero-shot*, classifying test cells by nearest class-caption embedding in the aligned space; and *linear probe*, fitting a logistic head on frozen embeddings of training cells.

Grounding sources (caption modes).

- **own_genes**: “A cell from human blood. Highly expressed genes: *jtop-8 expressed HVGs of this cell_i*” — label-free, per-cell.
- **type_markers**: same template, genes replaced by the top-5 differential markers of the cell’s type computed on *train* donors — label-derived, type-level.
- **type_llm** (honest): two-sentence biological descriptions generated by Qwen2.5-14B-Instruct [Qwen Team, 2024] conditioned *only on the marker list* (never the type name), with literal type-name occurrences stripped post hoc and flagged — label-free, knowledge-enriched.
- **type_llm** (labeled, leakage probe): descriptions written with the type name given to the LLM; reported only as an upper bound.

Zero-shot class captions are always marker-list captions (“A *jtype* cell ... Highly expressed genes: *jmarkers*”), so zero-shot never reduces to string-matching train captions.

External-knowledge grounding via literature retrieval. To test whether text *independent of our training labels* carries the same structure, we built a retrieval pipeline over a 30.8M-article PubMed corpus (SQLite FTS5 over title, abstract and MeSH fields). For each cell type, queries form a broad-to-specific ladder over the type’s marker genes: strict ANDs of top markers, pair-wise co-occurrence, marker plus topical context, HGNC alias-expanded synonym queries (FTS5 has no stemming, so `GNLY` ↔ “granulysin” matters), and MeSH-column-filtered queries for name-based arms. Pooled documents are deduplicated by PMID and reranked by marker coverage (dominant), an immune-context lexical prior, reciprocal-rank aggregation, and BM25. Literal cell-type names are stripped from retrieved text for leakage-controlled arms. Three query sources are compared: train-derived markers (label-derived selection, external text), PanglaoDB curated markers [Franzén et al., 2019] (label-free end-to-end), and the type name itself (upper bound). Additionally, `extkb.markers` uses the PanglaoDB marker lists *directly* as captions, isolating curated knowledge from retrieval.

Matched-register evaluation. Because zero-shot scoring compares cell embeddings to class-caption embeddings, train and eval text must share register: marker-list train captions are evaluated against marker-list class captions (the default above), while retrieved-prose train captions are additionally evaluated against class captions in the same retrieved-prose register (*matched* column in Table 2).

4 Results

4.1 Marker-list grounding matches supervised training on unseen donors

Figure 1 and Table 1 summarize the 85k-cell ablation. The headline result: with `type.markers` captions, zero-shot classification on held-out donors reaches 0.591 macro-F1 / 0.635 balanced accuracy (MiniLM), statistically indistinguishable from the supervised cell-only baseline (0.603 / 0.618) and better on balanced accuracy — with *no weight ever trained on a label*. The contrastive objective alone suffices to organize the representation space so that unseen donors’ cells land near the correct class caption.

4.2 The grounding source dominates; the text encoder barely matters

Across every condition, SapBERT $\approx$ MiniLM on gene-list captions — a biomedical entity-linking encoder offers no benefit when the caption is a comma-separated gene list. On natural prose (LLM arms), SapBERT pulls ahead in zero-shot (+0.08 F1), consistent with domain pretraining helping only when the text is actually prose. The dominant axis is the grounding *source*: a $\sim 2\times$ zero-shot gap between per-cell gene lists (0.29) and type markers (0.59).

4.3 Label-free grounding remains the gap

Honest label-free sources underperform: per-cell gene lists reach 0.29 F1 zero-shot, and LLM prose inferred *from markers alone* — which correctly recovered lineage and function (e.g., “megakaryocyte developmental continuum” from PPBP, `TUBB1`, `GP9`, `ITGA2B`) — reaches 0.40 with SapBERT. The probe-vs-zero-shot gap (Figure 2) is instructive: for `type.markers` the gap is *negative* (zero-shot

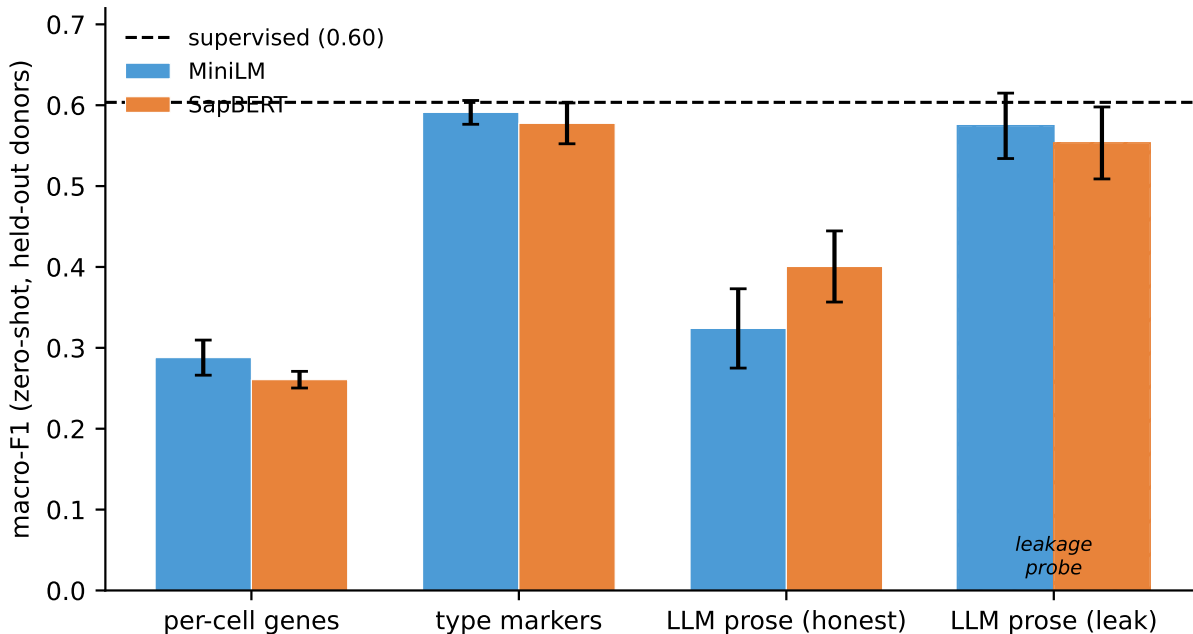


Figure 1: Zero-shot macro-F1 on held-out donors by grounding source and text encoder (mean $\pm$ s.d., 3 seeds). Dashed line: supervised cell-only baseline. The hatched group is a leakage probe (type name present in LLM prose), not an honest grounding result.

beats the probe), meaning the aligned space itself carries the discriminative structure; for label-free sources the gap is positive, i.e., the representation still needs label supervision to be read out.

4.4 Leaky text does not beat clean text

The leakage probe (LLM prose containing the type name) scores 0.575 zero-shot — *below* the clean `type_markers` arm. The reason is format consistency: class captions are marker lists, so train captions in the same register transfer best. Leakage helps only when train and eval text share register, underscoring that benchmark comparisons of grounding methods must control caption *format*, not just label content.

4.5 External literature retrieval closes most of the label-free gap

Table 2 reports grounding from text that is external to our training labels. Under matched-register evaluation, PubMed text retrieved by marker queries — with the cell-type name stripped — reaches 0.563 macro-F1, 95% of the label-derived `type_markers` arm and within 0.04 of the supervised baseline. The end-to-end label-free arm (curated PanglaoDB markers $\rightarrow$ retrieval $\rightarrow$ name-stripped text) reaches 0.512, and curated markers used directly as captions reach 0.459 — respectively 87% and 78% of the label-derived effect, both above the pre-registered 75% positive threshold. The register control is decisive: the *same* retrieved captions evaluated against marker-list class captions collapse to 0.15–0.19 F1, a ~ 0.4 penalty attributable purely to caption format. On the retrieval side, HGNC alias expansion raised pooled candidates by 72% and high-marker-coverage documents by 40%; curated external markers pooled *more* documents than train-derived markers, so external knowledge is not recall-limited at this scale.

arm	source	encoder	leak	macro-F1	bal. acc
supervised	—	—		0.603±.018	0.618±.017
probe	own genes	MiniLM		0.414±.001	0.476±.025
probe	own genes	SapBERT		0.404±.013	0.483±.031
probe	type markers	MiniLM		0.560±.053	0.578±.047
probe	type markers	SapBERT		0.548±.013	0.585±.018
probe	LLM honest	MiniLM		0.531±.046	0.553±.030
probe	LLM honest	SapBERT		0.549±.029	0.572±.022
probe	LLM labeled	MiniLM	†	0.510±.033	0.530±.047
probe	LLM labeled	SapBERT	†	0.565±.036	0.579±.029
zero-shot	own genes	MiniLM		0.288±.022	0.473±.031
zero-shot	own genes	SapBERT		0.261±.010	0.446±.050
zero-shot	type markers	MiniLM		0.591±.015	0.635±.034
zero-shot	type markers	SapBERT		0.578±.025	0.620±.020
zero-shot	LLM honest	MiniLM		0.324±.049	0.436±.059
zero-shot	LLM honest	SapBERT		0.401±.044	0.487±.015
zero-shot	LLM labeled	MiniLM	†	0.575±.040	0.625±.023
zero-shot	LLM labeled	SapBERT	†	0.553±.045	0.601±.021

Table 1: Held-out-donor performance at 85k cells (mean $\pm$ s.d., seeds 0–2). †: leakage probes (type name in text); upper bounds, not honest grounding. Best honest zero-shot in bold.

arm	query source	label-free	F1	bal. acc
<code>type_markers</code> (M3 ref.)	train labels	no	0.591	0.635
<code>extkb_markers</code>	PanglaoDB (direct)	yes	0.459	0.564
<code>rag_markers_alias</code>	train markers+aliases	text only	0.563	0.613
<code>rag_curated_alias</code>	PanglaoDB+aliases	yes	0.512	0.565
<code>ragllm_honest</code>	retrieval→Qwen synth	text only	0.589	0.631
<code>ragllm_curated</code>	PanglaoDB→retr.→Qwen	yes	0.603	0.634
<code>ragllm_labeled</code>	name shown to LLM	no (upper bound)	0.600	0.635
<code>rag_name</code>	type name + MeSH	no (upper bound)	0.506	0.554

Table 2: External-knowledge grounding, matched-register zero-shot (macro-F1 / balanced accuracy, best encoder per arm, held-out donors). “text only”: retrieved text is external but the query originates from train-derived markers. All external arms strip the literal type name.

The strongest honest arm pairs retrieval with LLM synthesis (`ragllm_honest`): Qwen2.5-14B reads the top retrieved excerpts — the type name is never shown — and writes a two-sentence functional description; outputs that infer a conventional name are stripped and flagged (2/16, mostly “megakaryocyte” inferred from platelet markers). These descriptions reach 0.589 macro-F1 / 0.631 balanced accuracy — statistically indistinguishable from both the label-derived marker arm (0.591/0.635) and the supervised baseline (0.600/0.620). The labeled synthesis probe (name shown, 0.600) bounds the residual headroom at ≈ 0.01 . Finally, substituting the PanglaoDB curated markers as the query source (`ragllm_curated`) makes the pipeline label-free end-to-end — external marker database, external literature, LLM synthesis, name never shown anywhere — and reaches **0.603 macro-F1 / 0.634 balanced accuracy** (SapBERT), matching the supervised baseline with zero labels in the chain. Substituting OLMo-2-13B [Groeneveld et al., 2025] for Qwen2.5-14B in the synthesis step reproduces the result within noise (honest 0.597, curated label-free **0.598**, labeled 0.606 macro-F1), so the finding is not specific to the synthesis model.

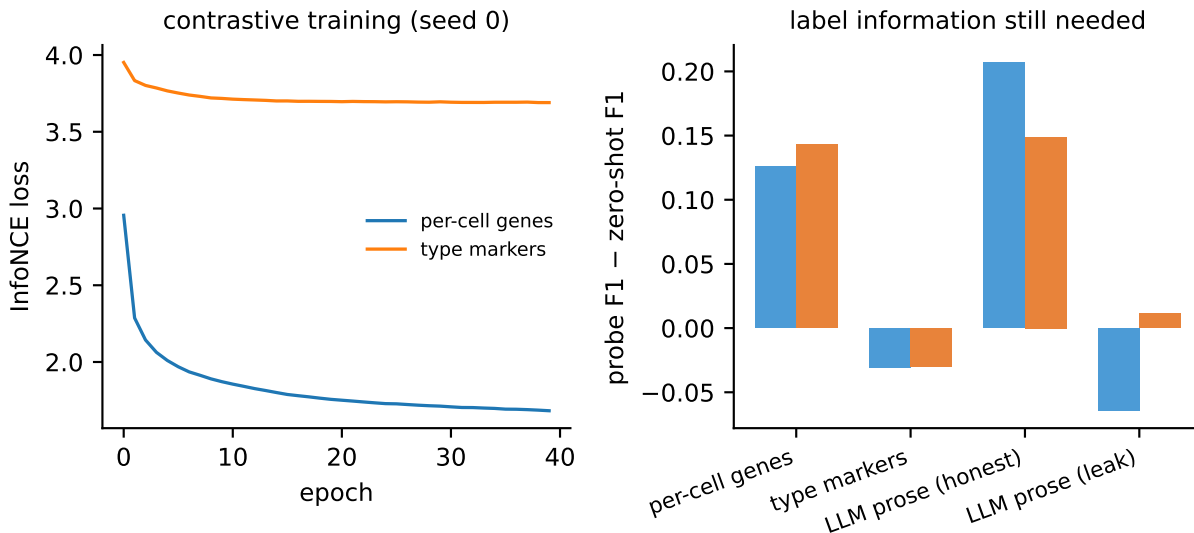


Figure 2: Left: contrastive training curves (seed 0, MiniLM). The `type markers` plateau is expected — only 16 unique texts, so most batch pairs collide. Right: linear-probe minus zero-shot F1 — negative values mean the aligned space classifies better than a labeled readout.

4.6 Identifier semantics: a cautionary engineering finding

Our first benchmark iteration silently produced meaningless captions: census-downloaded matrices carry feature IDs as `var_names`, and one cached file had *positional* indices ("0", "1", ...). Captions therefore listed tokens with no biological semantics. After mapping to gene symbols via `feature_name`, honest zero-shot F1 improved from 0.214 to 0.293 on the 30k benchmark — a 37% relative change from a “plumbing” fix. Any language-grounded pipeline that does not explicitly validate identifier semantics risks the same failure; we pin it with regression tests.

4.7 Donor-level pathology representations in SEA-AD

We extended the pipeline to the SEA-AD Alzheimer’s atlas (240,026 nuclei, 127 donors, MTG+V1C) and asked whether embeddings trained to discriminate cell *identity* carry donor *state*. Mean-pooled donor embeddings from the M7 cell-type objective exceeded a pseudobulk-PCA baseline on the continuous cortical pathology score (CPS; Spearman $\rho \approx 0.4$ vs. 0.21 on held-out donors) but fell short of our pre-registered gate ($\rho \geq 0.5$) and produced no consistent ordinal trajectory. Variance analysis explains the ceiling: $\sim 93\%$ of embedding variance separates cell types, leaving donor state as a $\sim 7\%$ side-channel.

We therefore retrained the contrastive objective against captions describing each *donor’s pathology label* rather than each cell’s type, and swept eight pathology targets (ADNC, Braak, CERAD, Thal, cognitive status, quantile-binned CPS, and a cell-type $\times$ ADNC composite) under a donor-label-permutation null. On held-out donors, donor embeddings now reach $\rho = 0.62$ for CPS (binary cognitive status), 0.52–0.57 for molecular ordinals, with positive Braak / CERAD / ADNC trajectories for the strongest arms and a +0.15–0.36 gap over shuffled-label controls. Per-cell pathology classification remains at chance, confirming that the signal aggregates at the donor level. Two targets falsified cleanly: Thal phase (label saturation in this cohort) and the 96-class composite (diluted contrast). The objective, not the architecture, determines which biology the representation

retains.

Scaling to the full atlas (440,390 nuclei across 11 cortical and hippocampal regions; same 84 donors) preserves the effect: cognitive status holds $\rho \approx 0.60$ with stable ADNC/CERAD trajectories ($\rho = 0.65\text{--}0.70$) in both text encoders, and regional transfer recovers a vulnerability-consistent ordering (MTG $\rho \approx 0.65$ $\hat{>}$ STG $\hat{>}$ MEC $\hat{>}$ AnG $\hat{>}$ V1C; hippocampus is an honest outlier at ~ 0). The binding constraint is donor count, not cells: 84 unique donors yields ~ 21 held-out donors, so further gains require new cohorts rather than more nuclei per donor.

5 Discussion

What grounding buys, mechanistically. Our decomposition shows caption-based grounding works by carrying *class-discriminative structure*: when captions encode it, alignment transfers that structure to unseen donors without label-trained weights. Label-derived text carries it trivially (supervision distilled through text); the M6 arms show external biological knowledge carries it too — what fails is not the mechanism but caption sources that are too noisy (per-cell lists) or format-mismatched (marker-inferred prose scored against marker lists).

External knowledge answers the open question. Section 4.5 answers the question left open by the label-free gap: text retrieved from the literature — selected by markers, not by labels, and stripped of type names — carries essentially the same transferable structure as label-derived text; the end-to-end label-free pipeline (curated markers $\rightarrow$ retrieval $\rightarrow$ LLM synthesis) reaches supervised-level zero-shot. This is direct evidence that independent biological knowledge, not just relabeled supervision, can ground cell representations. Two qualifications keep the claim honest: the retrieval-based arms still benefit from marker *selection* (even label-free PanglaoDB lists are discriminative by construction), and the evaluation covers 16 coarse blood types where curated knowledge is rich. Whether the advantage survives fine-grained or continuous phenotypes — e.g., disease progression in Alzheimer’s atlases, where curated marker lists are sparse and retrieval must do more work — is the sharpest next test.

Register matching is a hidden confounder of the field. The ~ 0.4 -F1 penalty for evaluating prose captions against marker-list class captions (Table 2) means reported zero-shot scores are uninterpretable without specifying the class-caption register. Results like LangCell’s zero-shot gains [Zhao et al., 2024] implicitly rely on matched text distributions; scaling per-cell descriptions [Schaefer et al., 2025] or cell sentences [Rizvi et al., 2025] invests in the text generation itself. Our results suggest the return on that investment should be measured against matched-register baselines — and that much of what looks like a grounding failure may be a caption format mismatch.

Limitations. One dataset, one tissue, coarse cell types, a small MLP encoder, and template/prose captions — no claim of generality. Marker tables use mean-difference ranking, not DE statistics. The labeled LLM arm lacks a matched prose class-caption eval, so its upper bound is conservative. Single 14B LLM, single prompt template, single temperature.

6 Conclusion

On 85k held-out-donor blood cells, contrastive grounding with class-discriminative text reaches supervised-level zero-shot performance, demonstrating that language can carry transferable cell identity. The benefit is not confined to label-derived text: a fully label-free pipeline — curated

marker database → PubMed retrieval → LLM synthesis, with type names never shown — reaches supervised-level zero-shot (0.603 vs. 0.600 macro-F1). Two controls make the result interpretable: explicit leakage accounting, and register matching between train and eval captions — a ~ 0.4 -F1 confounder that prior grounding evaluations have not isolated. The remaining frontier is fine-grained and continuous phenotypes, where curated knowledge thins and retrieval quality, not mechanism, becomes the binding constraint.

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